

SAFETY DATA SHEET



Product name n-Butyl acetate
MSDS number 80015
Revision Number 10***

AGHS/EN

Revision Date Apr.21.2016***
Issuing date Apr.21.2016***

1. Identification of the substance/preparation and the company/undertaking

Product name
n-Butyl acetate

Manufacturer or supplier's details

Celanese (Shanghai) International Trading Co., Ltd.

Room 239, Xinmao Building
South Taizhong Road
Waigaoqiao Free Trade Zone
Shanghai, China

Celanese Pte Ltd

138 Robinson Road
#17-00
Singapore (068906)

Product Information

HazCom@celanese.com***

Emergency telephone number

+(65) 62656917 (Operations Room direct dial)
or fax request to +(65) 62664696 (Facsimile to Operations Room)
or email to posh.er@paccoffshore.com.sg

In China Emergency Number: 86-532-83889090 (NRCC)

Identified uses

Solvent

2. Hazards identification

GHS Classification

Hazards

Flammable liquid
Specific target organ systemic toxicity (single exposure)
Acute aquatic toxicity

Category

Category 3
Category 3
Category 3

Symbol(s)



Labeling

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Signal Word Warning

Hazard Statements
H226 - Flammable liquid and vapor
H336 - May cause drowsiness and dizziness
H402 - Harmful to aquatic life

Precautionary Statements
P210 - Keep away from heat/sparks/open flames/hot surfaces. - No smoking
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower
P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
P312 - Call a POISON PHYSICIAN/doctor/physician if you feel unwell.
P273 - Avoid release to the environment

3. Composition/Information on ingredients

Chemical characterization Acetic acid n-butyl ester

Components	CAS-No	Percent %
n-Butyl acetate	123-86-4	min 99.5

4. First aid measures

General Information

Remove contaminated, soaked clothing immediately and dispose of safely. Pay attention to own protection. In any case show the physician the Safety Data Sheet.

Skin

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. If symptoms persist, call a physician.

Eyes

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Call a physician immediately.

Inhalation

Keep at rest. Move to fresh air. Call a physician immediately.

Ingestion

Rinse with plenty of water. If swallowed, do not induce vomiting - seek medical advice.

5. Fire-fighting measures

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5. Fire-fighting measures

NFPA: Health: 2

Flammability: 3

Instability: 0

Suitable extinguishing media

Foam, Dry chemical, Carbon dioxide (CO₂)

Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

Special exposure hazards arising from the substance or preparation itself, its combustion products, or released gases

Under conditions giving incomplete combustion, hazardous gases produced may consist of

Carbon monoxide

Carbon dioxide (CO₂)

Combustion gases of organic materials must in principle be graded as inhalation poisons

Vapors are heavier than air and may spread along floors

Special protective equipment for fire-fighters

self-contained breathing apparatus (EN 133).

Environmental precautions

Water used to fight fire runoff can cause environmental damage. Dike and collect water used to fight fire.

Other Information

Cool containers / tanks with water spray.

6. Accidental release measures

Personal precautions

Avoid contact with the skin and the eyes. Keep away from heat and sources of ignition. Provide adequate ventilation.

Environmental precautions

Prevent further leakage or spillage. Do not discharge into the drains/surface waters/groundwater.

Methods for cleaning up

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Dispose of in accordance with local regulations***

7. Handling and storage

Advice on safe handling

Provide sufficient air exchange and/or exhaust in work rooms.

Incompatible products

Keep away from: strong acids, peroxides, oxidizing agents, amines

Protection - fire and explosion:

Keep away from sources of ignition - No smoking. Take necessary action to avoid static electricity discharge.

Ground and bond containers when transferring material. In case of fire, emergency cooling with water spray should be available.

Material storage

Store locked up. Keep in a dry, cool and well-ventilated place.

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Incompatible products

Keep away from: strong acids, peroxides, oxidizing agents, amines

Technical measures/Storage conditions

Keep tightly closed in a dry, cool and well-ventilated place. Handle an open container with care.

8. Exposure controls / personal protection

ACGIH Exposure Limits

Components	TWA
n-Butyl acetate	150 PPM

Components	STEL
n-Butyl acetate	200 PPM

OSHA Exposure Limits

Components	TWA
n-Butyl acetate	150 PPM

Components	STEL
n-Butyl acetate	200 PPM

Exposure controls

Engineering measures

General or dilution ventilation is frequently insufficient as the sole means of controlling employee exposure. Local ventilation is usually preferred. Explosion-proof equipment (for example fans, switches, and grounded ducts) should be used in mechanical ventilation systems.

Personal protective equipment

General advice

Avoid contact with skin and eyes. Do not breathe vapors or spray mist. Use only in an area equipped with a safety shower. Hold eye wash fountain available.

Hygiene measures

When using, do not eat, drink or smoke. Take off all contaminated clothing immediately. Wash hands before breaks and immediately after handling the product..

Respiratory protection

If aerosols or vapors are present, respiratory protection is required (gas filter A) .

Eye protection

Tightly fitting safety goggles. In addition to goggles, wear a face shield if there is a reasonable chance for splash to the face. Equipment should conform to EN 166.

Skin protection

impervious clothing

Hand protection

Chemicals resistant gloves

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Suitable material	Butyl-rubber
Type	Butoject (Company KCL) or comparable article; or refer to glove manufacturer's recommendation
Evaluation	according to EN 374: level 3
Material thickness	approx. 0.7 mm
Break through time	approx. 60 min

9. Physical and chemical properties

Appearance

Form	liquid
Color	colourless
Odor	fruity

Odor Threshold 7 - 20 ppm (gas in air)

Molecular Weight 116.16 g/mol

Flash point 24°C

Method DIN EN ISO 13736

Ignition temperature 415°C

Method DIN 51794

Decomposition Not determined***

Temperature

Lower explosion limit 1.2 Vol. %

Upper explosion limit 7.5 Vol. %

Flammability (solids) not applicable

Melting point/range -78°C

Boiling point/range 126.1°C @ 1013 hPa

Method DIN 53171

Density 0.881 g/ml @ 20°C

pH 6.2 @ 20°C

Concentration @ 5 g/l

Viscosity 0.685 mPa*s @ 25°C

Vapor pressure 15 hPa @ 20°C

63 hPa @ 50°C

Vapor density 4.0 (Air=1)

Evaporation Rate 1.0 (n-Butyl acetate = 1)

Water solubility 5.3 g/l @ 20°C

Solubility in other solvents miscible with, Ethanol, Diethyl ether, soluble in, Acetone, Chloroform

Partition coefficient 2.3 (measured)

(n-octanol/water)

Explosive Properties not applicable based on consideration of the structure

Oxidizing Properties not applicable based on consideration of the structure

Surface Tension 61.3 mN/m @ 20°C

Concentration 1 g/l

Dissociation constant not applicable based on consideration of the structure

10. Stability and reactivity

Chemical stability

Stable under normal conditions of handling, use and transportation.***

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Conditions to avoid

Avoid any source of ignition. Avoid contact with heat, sparks, open flame, and static discharge.***

Incompatible Materials

Keep away from:., peroxides, oxidizing agents, strong acids, amines

Hazardous Combustion or Decomposition Products:

Thermal decomposition products may include oxides of carbon.

11. Toxicological information

Potential health effects

Routes of exposure Skin, eyes, inhalation.

Immediate effects

Skin Prolonged or repeated contact may dry skin and cause irritation. Symptoms of overexposure include: Drying, cracking or inflammation of skin.***

Eyes Essentially non-irritating.

Inhalation Vapors may cause drowsiness and dizziness

Ingestion Essentially non-toxic.

n-Butyl acetate

Acute oral toxicity LD50: 10760 mg/kg
Acute dermal toxicity LD50: > 14000 mg/kg
Skin corrosion/irritation No skin irritation
Species rabbit
Method OECD 404

Skin Sensitization nonsensitizer
Species guinea pig
Method OECD 406

Serious eye damage/eye irritation No eye irritation
Species rabbit eye
Method OECD 405

Carcinogenic effects No data

in vitro Mutagenicity Ames Test: negative - with and without metabolic activation -
Method: OECD 471 Cell gene-mutation in Chinese Hamster
Cells: negative - with and without metabolic activation -
Method: OECD 476 (Reference substance: n-Butanol)
Chromosome aberrations in Chinese Hamster Cells:
negative - without metabolic activation - Method: OECD 473
in vivo Mutagenicity Mammalian Erythrocyte Micronucleus Test in mice: negative
- Method: OECD 474 (Reference substance: n-Butanol)

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Reproductive toxicity	No effects on fertility
Routes of exposure	inhalation
Species	rat
	NOAEC: 9640 mg/m ³
Developmental effects	No teratogenic, maternal or developmental effects
Routes of exposure	Inhalation
Species	rat
	LOAEC: 7230 mg/m ³
Neurotoxicity	No evidence of neurotoxicity
Type of study	6-hour inhalation toxicity study
Routes of exposure	inhalation
	NOAEC: 2410 mg/m ³

12. Ecological Information

n-Butyl acetate

Acute fish toxicity	LC50: 18 mg/l (96h)
Species:	Pimephales promelas (Fathead minnow)
Method	OECD 203
Acute daphnia toxicity	EC50: 44 mg/l (48h)
Species:	Daphnia magna
Toxicity to aquatic plants	EC50: 648 mg/l (72h)
Species:	Desmodesmus subspicatus
Biodegradation	Readily biodegradable
	83 % (28d)
Method	OECD 301 D
Other potential hazards	The substance does not meet the criteria for PBT / vPvB according to REACH, Annex XIII

13. Disposal considerations

Product information

Disposal required in compliance with all waste management related state and local regulations. The choice of the appropriate method of disposal depends on the product composition by the time of disposal as well as the local statutes and possibilities for disposal

Uncleaned empty packaging

Contaminated packaging should be emptied as far as possible and after appropriate cleansing may be taken for reuse

14. Transport information

US Department of Transportation

UN/NA Number:	UN 1123
Proper Shipping Name	Butyl acetates
Hazard class	3
Packing Group	III
Reportable Quantity (RQ)	5000lb/2270kg
Emergency Resp. Guide	128

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14. Transport information

ADR/RID

UN/ID No. UN 1123
Proper Shipping Name Butyl acetates
Hazard Class 3
Classification Code F1
Packing group III
Environmentally hazardous no
Tunnel Restriction Code (D/E)
Hazard Label(s) 3
Hazard Number 30

ADN

ADN: Container and Tanker

UN/ID No. UN 1123
Proper Shipping Name Butyl acetates
Hazard Class 3
Classification Code F1
Packing group III
Environmentally hazardous no
Hazard Labels 3

ICAO/IATA

UN-No. UN 1123
Proper Shipping Name Butyl acetates
Hazard Class 3
Packing group III
Environmentally hazardous no
Hazard Labels 3

IMDG

UN/ID No. UN 1123
Proper Shipping Name Butyl acetates
Hazard Class 3
Packing group III
Marine pollutant no
Hazard Labels 3
EmS Code F-E, S-D

15. Regulatory information

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15. Regulatory information

International Inventories

Listed on the chemical inventories of the following countries or qualifies for an exemption:

Australia (AICS)
Canada (DSL)
China (IECSC)
Europe (EINECS)
Japan (ENCS)
Japan (ISHL)
Korea (KECI)
New Zealand (NZIoC)
Philippines (PICCS)
United States (TSCA)

16. Other information

HMIS: Health: 2 Flammability: 3 Physical Hazard: 0

Prepared By

Product Stewardship Department
Celanese

Other Information:

Observe national and local legal requirements.

Changes against the previous version are marked by ***

Sources of key data used to compile the datasheet

Information contained in this safety data sheet is based on Celanese owned data and public sources deemed valid or acceptable. The absence of data elements required by ANSI or 1907/2006/EC indicates that no data meeting these requirements is available.

Further information

This information is based on our present state of knowledge. It shall describe our products regarding safety requirements and shall not be construed as a guarantee or statement of condition and/or quality. For more information, other material safety data sheets or technical data sheets please consult the Celanese homepage (www.celanese.com)